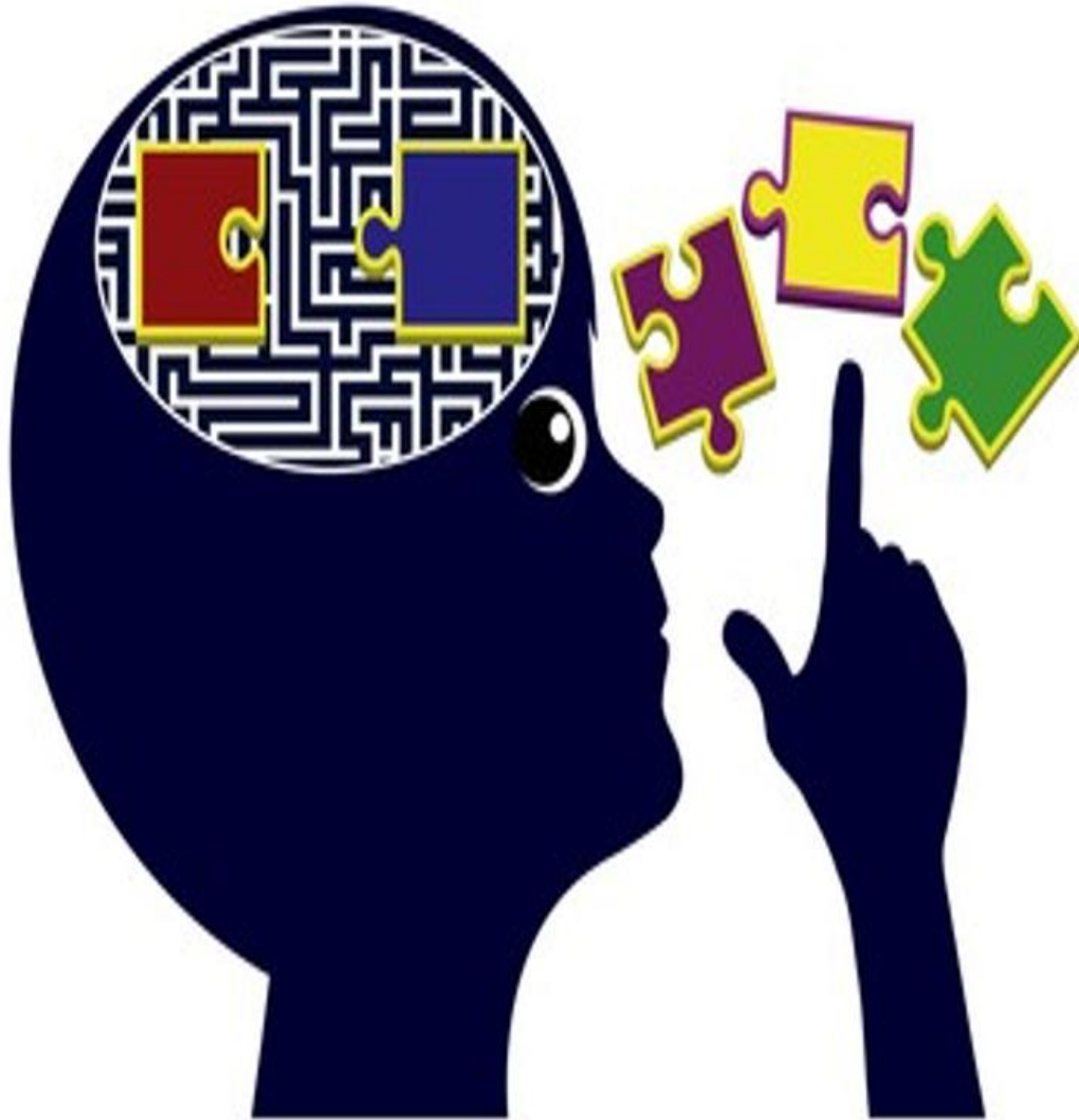




Critical Thinking And Logic

Lesson 1: Basic Concepts of
Logic: Arguments, Premises and
Conclusions

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What is the Meaning of Logic ?

- Logic is a science that evaluates arguments.
- Logic is the study of methods for evaluating whether the premises of arguments adequately support or provide a good evidence for the conclusions.



Relation Between Critical Thinking and Logic

- Logic and Critical Thinking are essentially linked to each other.
- Logical thinking is a useful set of tools ideally used within critical thinking.
- Critical thinking is a process of evaluation which uses logic to separate truth from falsehood, reasonable from unreasonable beliefs.



Major Benefits From The Study Of Logic

1. It helps us to distinguish good arguments from bad arguments.
2. It helps us to understand and identify the common logical errors in reasoning.
3. It helps us to understand and identify the common confusions that often happen due to misuse of language.



What is an Argument ?

- **An argument :**

It is a group of statements , statement consists of two main parts: premises and conclusion. Statement may be true or false.

- **Premises:**

Are the statements that provide evidence, reasons, or assumptions for the conclusion.

Argument = premises + conclusion

- **Conclusion:**

Is the statement that follows from the premises and expresses the main point or claim of the argument

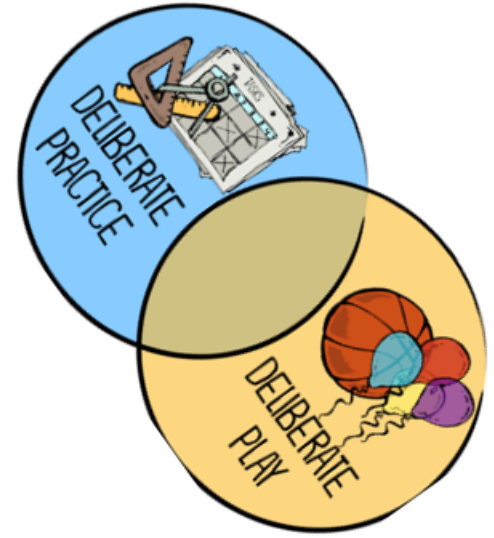
Let's practice and play

There are **sentences** that are **not statements**, find them :

1. *Abd El Fattah El Sisi Is The President of Egypt .*
2. *Would you close the window?*
3. *Cairo is the capital of Egypt .*
4. *Ethiopia was colonized by Germany .*
5. *Give me your ID Card, Now!*
6. *Let us study together. (Proposal)*



1,3- true statement 2-question 4- false statement 5- command 6-proposal





Let Us Now **Construct Arguments** Together

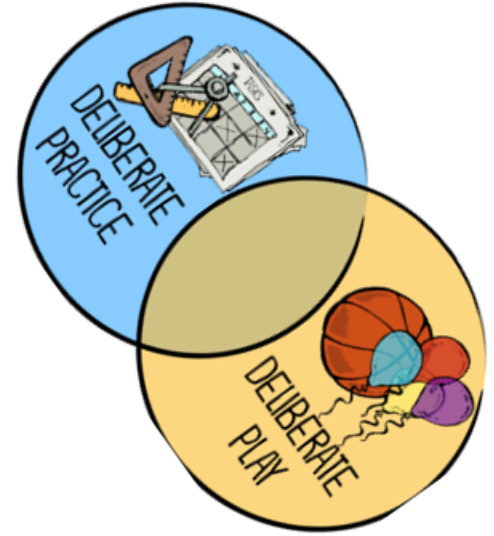
Argument 1		Argument 2	
<i>All Ethiopians are Africans.</i>	<i>(Premise 1)</i>	<i>Some Africans are black.</i>	<i>(Premise 1)</i>
<i><u>Tsionawit</u> is Ethiopian.</i>	<i>(Premise2)</i>	<i><u>Zelalem</u> is an African.</i>	<i>(Premise 2)</i>
<i>Therefore, <u>Tsionawit</u> is African.</i>	<i>(Conclusion)</i>	<i>Therefore, <u>Zelalem</u> is black.</i>	<i>(Conclusion)</i>

List of premises and conclusion key words

Premises key words	Conclusion key words
<i>Since</i> <i>As indicated by</i> <i>Because</i> <i>Owing to</i> <i>Seeing that</i> <i>Given that</i> <i>As</i> <i>For</i> <i>In that</i> <i>May be inferred from</i> <i>Inasmuch as</i> <i>For the reason that</i>	<i>Therefore</i> <i>Wherefore</i> <i>Accordingly</i> <i>Provided that</i> <i>It must be that</i> <i>We may conclude</i> <i>so</i> <i>Hence</i> <i>It shows that</i> <i>Whence</i> <i>Thus</i> <i>Consequently</i> <i>We may infer</i> <i>It implies that</i> <i>As a result</i> <i>and</i> <i>It follows that</i>

Let's practice and play

Which of the following represents **Arguments**:

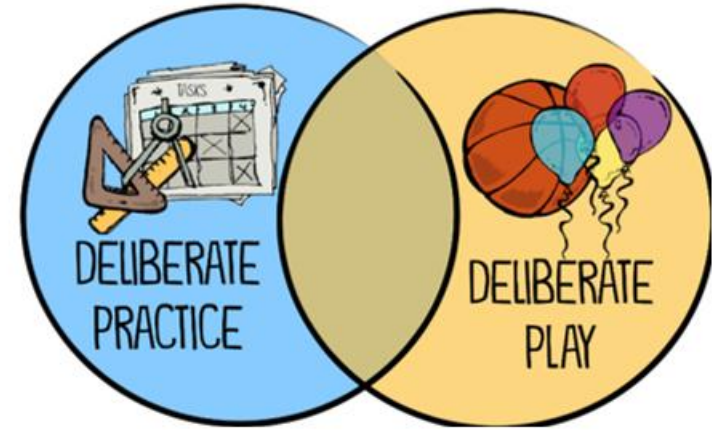


I followed the directions when cooking meat, and it worked for me.



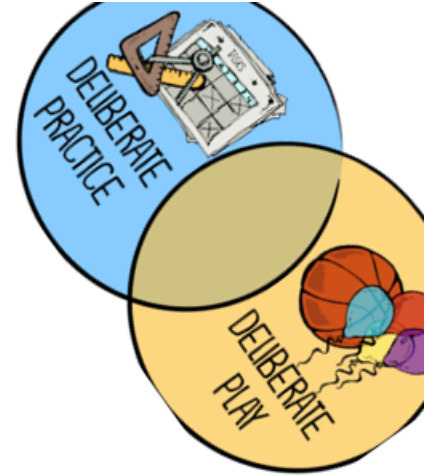


Answer



This is **not** an argument. It directly reports experience and doesn't try to convince.

Let's Practice and Play

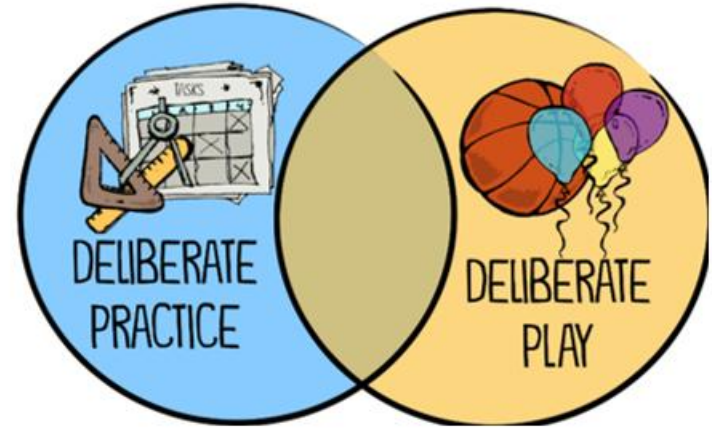


- Which of the following represents **Arguments**:
- No scientific hypothesis can be confirmed because no evidence we find could rule out the possibility of contrary evidence in the future.





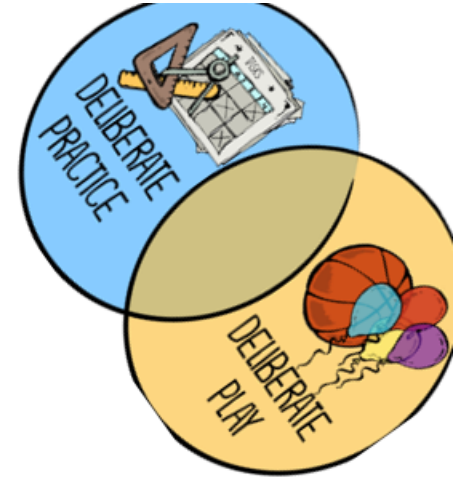
Answer



This is **an argument**. The conclusion is: “No scientific hypothesis can be conclusively confirmed.”

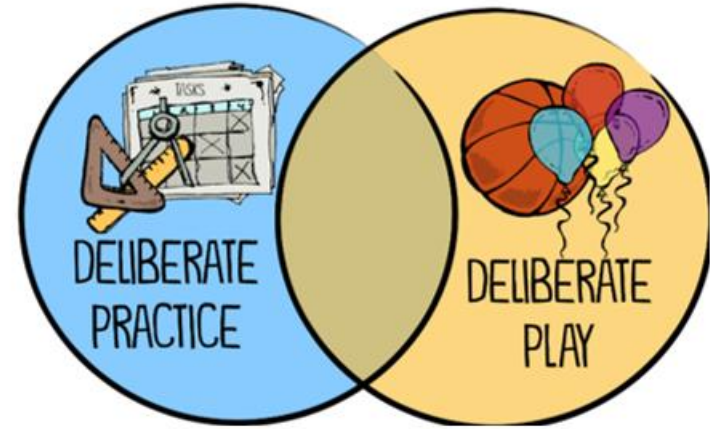
Let's Practice and Play

- Which of the following represents **Arguments**:
- Studying philosophy helps us to excel in many other areas.
Therefore, studying philosophy is a worthwhile .





Answer



its is **an argument** Notice how the conclusion indicator expression “**therefore**” helps us to identify the inference in this argument.